Industry Problem Solving Task

1. Network Design for a 20-Person Startup

Recommended Topology: Star Topology

A **star topology** is the most suitable network design for a small startup with around 20 employees. In this topology, all computers and network devices are connected to a central switch. The central switch controls communication between the connected devices.

Network Components

- A **24-port managed Layer 2/Layer 3 switch** can be used as the central device.
- Computers, printers, IP phones, and other devices are connected to the switch.
- **Cat6 UTP cables** are used because they support Gigabit Ethernet and are suitable for normal office distances.
- A **router/firewall** connects the internal network to the Internet.
- A **wireless access point** provides Wi-Fi connectivity for laptops, smartphones, and tablets.
- DHCP can be configured on the router or server to automatically assign IP addresses.

Working

When one employee wants to communicate with another employee or access the Internet, the data first reaches the central switch. The switch forwards the data to the correct destination. Internet traffic is sent from the switch to the router/firewall and then to the Internet.

Advantages

1. Easy to install and manage.
2. Easy to identify and troubleshoot failures.
3. Failure of one cable or computer does not affect other users.
4. Easy to add new computers and devices.
5. Provides good network performance.
6. Centralized management improves security and monitoring.

Disadvantage

The main disadvantage is that if the **central switch fails**, all connected devices lose network connectivity.

Justification

For a 20-person startup, star topology provides a good balance between **cost, performance, reliability, security, and scalability**. It is much easier to maintain than complex topologies.

Conclusion

Therefore, a **star topology with a managed switch, router/firewall, Wi-Fi access point, and Cat6 cables** is an effective solution for a small startup office.

2. Network Design for a Bank

Recommended Topology: Full or Partial Mesh Topology

A bank handles sensitive financial transactions and requires continuous network availability. Therefore, **full mesh or partial mesh topology** is suitable for the core banking network.

Network Components

- Multiple **enterprise routers** are used for redundancy.
- Core Layer 3 switches connect important servers and network devices.
- **Fiber-optic cables** are used for high-speed core connections.
- Firewalls protect the banking network from unauthorized access.
- UPS systems and backup power supplies provide continuous operation.
- Servers can use redundant storage such as RAID.
- Routing protocols such as **OSPF and BGP** can provide automatic path selection and failover.

Working

In a mesh network, important devices have multiple connections to other devices. If one link becomes unavailable, network traffic can use another available path.

For example, if Router A cannot communicate with Router B because of a cable failure, traffic can be redirected through another router. This prevents complete network failure.

Advantages

1. Very high reliability.
2. Provides multiple backup paths.
3. Minimizes network downtime.
4. Supports automatic failover.
5. Suitable for critical financial transactions.
6. Fiber provides high bandwidth and low latency.
7. Improves fault tolerance.

Disadvantages

1. Expensive to install.
2. Requires more cables and networking equipment.
3. Network configuration is more complex.
4. Requires skilled network administrators.

Justification

Banks cannot afford long network outages because customers depend on services such as ATM transactions, online banking, fund transfers, and payment systems. A mesh design provides **redundancy and fault tolerance**, making it appropriate for banking environments.

Conclusion

A **partial mesh at the core with redundant routers, switches, firewalls, fiber connections, and backup power** provides a highly reliable network for a bank.

3. Network Design for a Small Classroom Lab

Recommended Topology: Bus Topology

For a small classroom laboratory with a limited budget, a **bus topology** can be considered because all computers share a common communication medium.

Network Components

- A common backbone cable is used to connect computers.
- Traditional bus networks use **coaxial cable**.
- T-connectors connect individual computers to the backbone.
- Terminators are placed at both ends of the cable to prevent signal reflection.
- Network interface cards are required in the computers.
- No dedicated central switch is required in the traditional bus design.

Working

When one computer sends data, the data travels along the common backbone cable. All computers receive the signal, but only the computer for which the data is intended processes it.

For example, if Computer A sends a file to Computer C, the signal travels through the shared cable and Computer C accepts the data.

Advantages

1. Low installation cost.
2. Requires less networking equipment.

3. Uses less cable compared with some other designs.
4. Simple for a small network.
5. Suitable for basic classroom activities.
6. No dedicated server is necessary for basic file sharing.

Disadvantages

1. Failure of the backbone cable can affect the entire network.
2. Troubleshooting can be difficult.
3. Network performance decreases as more devices are added.
4. Collisions can occur in traditional shared-medium networks.
5. Limited scalability.
6. Not suitable for large or high-traffic networks.

Justification

A classroom with a small number of computers and low network traffic may not require an expensive enterprise network. Therefore, a simple topology can reduce the initial cost.

However, in a **modern real-world classroom**, a star topology using an inexpensive unmanaged switch is generally preferred because it provides better performance and easier troubleshooting.

Conclusion

Bus topology can be considered for a **very low-cost, small classroom environment**, but modern Ethernet networks normally use star topology because of its better reliability and performance.

4. Network Design for a Large University Campus

Recommended Topology: Hybrid Topology

A university campus contains many buildings, including classrooms, laboratories, libraries, hostels, administrative offices, and research centers. Therefore, a single topology is not sufficient. A **hybrid topology** is the most suitable solution.

Network Architecture

The university can use a **three-layer hierarchical architecture**:

1. **Core Layer**
2. **Distribution Layer**
3. **Access Layer**

Core Layer

The core layer forms the high-speed backbone of the campus.

- High-speed **fiber-optic cables** connect the major buildings.
- Ring or partial mesh topology can be used.
- Core Layer 3 switches and routers provide high-speed communication.
- Redundant connections provide backup paths.

Distribution Layer

Each building can have one or more distribution switches.

- Distribution switches connect the building to the campus core.
- Dual fiber uplinks can be provided for redundancy.
- VLANs can be used to separate different departments or services.

Access Layer

The access layer connects end-user devices.

- Computers, printers, IP phones, and other devices connect to access switches.
- **Cat6 Ethernet cables** can be used within buildings.
- Wireless access points provide Wi-Fi connectivity.
- PoE switches can supply power to access points and IP phones.

Advantages

1. Highly scalable for thousands of users.
2. Easy to add new buildings.
3. Provides high-speed communication.
4. Redundant links improve reliability.
5. Different departments can be separated using VLANs.
6. Centralized management is possible.
7. Supports both wired and wireless devices.

Disadvantages

1. High initial installation cost.
2. Requires professional network planning.
3. Fiber installation can be expensive.
4. Network management can be complex.

Justification

A university continuously expands its infrastructure and user base. A hybrid design allows each building to operate efficiently while remaining connected to the main campus backbone.

Conclusion

A hybrid topology with fiber-based mesh/ring backbone, star topology within buildings, and hierarchical core-distribution-access architecture provides the required scalability, reliability, performance, and flexibility.

5. Network Design for a Company with Multiple Departments

Recommended Design: Hierarchical Network with VLANs

A company containing departments such as **HR, Finance, Sales, Marketing, and IT** requires a secure and organized network. A hierarchical network using **VLANs and Layer 3 switching** is suitable.

Network Structure

The network can be divided into three major layers:

- **Access Layer:** Connects computers, printers, IP phones, and other user devices.
- **Distribution Layer:** Connects department switches and performs routing and policy control.
- **Core Layer:** Provides fast communication between different parts of the organization.

VLAN Configuration

Each department can be assigned a separate VLAN.

For example:

- VLAN 10 – HR
- VLAN 20 – Finance
- VLAN 30 – Sales
- VLAN 40 – IT
- VLAN 50 – Management

VLANs logically separate departments even if they use the same physical network infrastructure.

Working

A computer in the Finance department communicates with other Finance computers through the Finance VLAN. If it needs to communicate with the HR department, the traffic is sent to a **Layer 3 switch or router**, which performs inter-VLAN routing.

A firewall controls traffic between the internal network and the Internet and can also apply security policies.

Security

- VLANs reduce unnecessary broadcast traffic.
- Access control lists (ACLs) can restrict communication between departments.
- Firewalls protect the network from external threats.
- Strong authentication can be used for administrative access.
- Network monitoring can identify suspicious activity.

Advantages

1. Better network security.
2. Reduces broadcast traffic.
3. Improves network performance.
4. Easy to manage different departments.
5. Supports future expansion.
6. Provides controlled communication between departments.
7. Redundant links can improve reliability.
8. Centralized monitoring and management are possible.

Disadvantages

1. VLAN configuration requires technical knowledge.
2. Layer 3 switches and firewalls increase the cost.
3. Incorrect configuration can cause connectivity problems.
4. Regular maintenance and monitoring are required.

Justification

Different departments may have different security requirements. For example, Finance may handle confidential financial information, while HR manages employee records. VLANs and firewall policies can isolate sensitive departments and allow only authorized communication.

Conclusion

A hierarchical network using VLANs, Layer 2 access switches, Layer 3 switches, routers, and firewalls provides a secure, efficient, reliable, and scalable solution for a company with multiple departments.